Jawad Khan

Junior Data Scientist
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EDUCATION

University of Engineering and Technology (UET), Peshawar 2024 – Present (Expected 2028)
BS Data Science

INTERNSHIP EXPERIENCE

AI Intern — DevSpire Solutions Nov 2025 – Feb 2026

- Conducted data preprocessing and feature engineering on real-world datasets to support ML pipelines.
- Built and evaluated supervised ML models for classification and regression tasks using Python and scikit-learn.
- Contributed to a fraud detection project applying imbalanced classification techniques (Recall-focused evaluation).

Data Science Intern — Digital Empowerment Network (DEN) Jul 2025 – Sep 2025

- Performed end-to-end data analysis and supervised ML model development on assigned business datasets.
- Applied Logistic Regression and Decision Tree classifiers; delivered structured reports to cross-functional team.

PROJECTS

Financial Fraud Detection System
Python | Pandas | Scikit-learn | XGBoost | SMOTE | Machine Learning

- Developed an end-to-end fraud detection pipeline on a 1M+ transaction dataset, performing data exploration, preprocessing, feature analysis, and class-imbalance handling.
- Compared Logistic Regression, Random Forest, and XGBoost models using Precision, Recall, F1-score, and ROC-AUC.
- Analyzed transaction and customer-level patterns to identify factors associated with fraudulent activity and support data-driven risk decisions.

Online Retail Customer & Market Basket Analysis
Python | Pandas | Scikit-learn | SQL | Apriori | FP-Growth | K-Means | PCA

- Analyzed transactional retail data to uncover customer behavior, purchasing patterns, product relationships, and business opportunities.
- Applied RFM analysis, K-Means clustering, PCA, Apriori, and FP-Growth to segment customers and identify frequently purchased product combinations.
- Translated analytical findings into business-focused insights for customer segmentation, product recommendations, and sales strategy.

Air Quality Prediction System
Python | XGBoost | FastAPI | Streamlit | Machine Learning

- Developed an XGBoost regression model for AQI prediction using pollutant measurements and temporal features including lagged AQI values.
- Built a FastAPI inference service and interactive Streamlit application to expose the trained model through a user-facing prediction workflow.
- Deployed the application online, demonstrating an end-to-end workflow from data preprocessing and model development to API integration and deployment.

AI Data Science Project Architect
Python | LLMs | Hugging Face | Streamlit | Pydantic | Prompt Engineering

- Built an AI-powered application that converts user requirements into structured Data Science project blueprints.
- Designed workflows generating dataset recommendations, research papers, GitHub repositories, technology stacks, project requirements, and learning roadmaps.
- Implemented a modular Python architecture using Pydantic schemas, LLM integration, configuration management, and an interactive Streamlit dashboard.

Business Intelligence & Data Analytics
Power BI | Excel | SQL | Data Analysis | Data Visualization

- Developed interactive dashboards for retail and business datasets to monitor KPIs, sales performance, customer behavior, and operational trends.
- Performed data cleaning, exploratory analysis, KPI development, and interactive visualization to transform raw data into actionable business insights.

CERTIFICATIONS

Python for Data Science, AI & Development — IBM (Coursera) Jun 2025
Data Analysis with Python — IBM (Coursera) May 2025
Databases and SQL for Data Science with Python — IBM (Coursera) Feb 2025

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, Excel
Data Science & Machine Learning: Scikit-learn, XGBoost, Classification, Regression, Clustering, PCA, Feature Engineering, Model Evaluation
Analytics & Statistics: Exploratory Data Analysis, Hypothesis Testing, Statistical Analysis, RFM Analysis, Cohort Analysis, Business Analytics
Data Visualization & BI: Power BI, Matplotlib, Streamlit
AI & Application Development: LLMs, Hugging Face, LangChain, Ollama, FastAPI, REST APIs, Pydantic